Deep Experimental Investigation of NBTI Impact on CMOS Inverter Reliability

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***Abstract*—Negative Bias Temperature Instability has become a major reliability concern in CMOS circuits. This is due to the fact that PMOS transistors of these circuits are submitted to many parameters shifts resulting in circuit performance degradations. In order to mitigate NBTI, IC designers need to accurately evaluate the NBTI induced performance degradations of their circuits to conduct aging simulation of their circuit lifetime. For this purpose, we present in this work a deep experimental investigation of NBTI impact on the CMOS inverter. The experimental setup of the DC NBTI has been conducted by Measure Stress Measure Method where a set of Negative stress Voltages at different temperatures were continuously applied for duration of 3600s. The results show that NBTI does shift the inverter's Voltage Transfer Curve, decreases its logic threshold, unbalance its logic states, and degrades its noise immunity. The obtained results demonstrate that these degradations worsen with both high negative stress voltage and elevated temperature and consequently impact the CMOS inverter robustness and thus its reliability.**

***Index Terms*— NBTI stress, CMOS inverter, Noise Margins, Robustness. IC reliability.**

I. INTRODUCTION

The Negative Bias Temperature Instability, as a degradation phenomenon occurring mainly in MOS Field Effect Transistors MOSFETs [1] [2], is one of the main reliability problems of CMOS Integrated Circuits that limit considerably their life time [3] [4]. It is well known that this phenomenon is ascribed to Si/SiO_2 interface states and positive charges in the oxide resulting from the breaking down of Si-H bonds at the SiO_2/Si interface of PMOS transistors under negative Bias at elevated temperatures [2]. NBTI is characterized by a positive shift in the absolute value of the PMOS threshold voltage |Vthp| leading to the decrease of drain current I_{DS} and consequently to performances degradation of CMOS Integrated Circuits [5].

At device level, many research works have been performed to study the fundamentals behind NBTI effect on PMOS transistor parameters under both DC and AC stresses and to propose models for both degradation and recovery and lifetime prediction [6] [7] [8]. Nevertheless, the focus on the low-level details does not provide insight about the circuit degradation due to NBTI on a larger scale. This is due to the fact that PMOS transistors within a circuit are interacting with other PMOS and NMOS transistors by sharing with them power supply, and input and output signals and interconnects. So, to properly estimate NBTI degradation and carry out ageing simulation, a model that faithfully reflects the impact of NBTI on circuit performances is required. To obtain such model, an experimental NBTI characterization at circuit level is needed. In [9] an experimental characterization of NBTI has been done on CMOS inverters using stress/measure/tress (SMS) protocol conducted with one negative gate voltage at one temperature 125°C. In contrast with this work we have deeply investigate NBTI influence on CMOS inverter DC performances by applying a series of negative voltages and heating samples at different temperatures to accurately track their influence and to detect where they are more influencing the degradation.

The rest of the paper is organized as follows. In section II, the experimental DC NBTI setup details, samples, and SMS protocol are described. The obtained results are analyzed and discussed in section III. Section IV concludes the paper.

II. EXPERIMENTAL SETUP

The samples under test are CMOS inverters with aspect ratios of 1µm/0.8µm and 2µm/0.8µm for NMOS and PMOS transistors respectively. They were fabricated at ISiT (Institute for Silicon Technology) of Fraunhofer, Germany using N-well process. The gate oxide thickness is 20 nm.

As the time delay between the application of stress and measurement of degradation is quite important as it can produce a wide variation in the results, the SMS sequences have been performed using fully automated benches. The bench includes a sensitive Agilent HP 4156C for voltage measurements and trigging. A hotplate is used, inside Karl Suss PA300 micro-manipulator probe station, to vary the temperature of the chip during the experiments. The test circuit chip (non-packaged) within and, the probe station were isolated from vibration and enclosed in a grounded faraday cage to avoid both RF and light effects.

The test protocol we adopted is described as follows: during stress period of 3600s, a negative voltage was applied to the inverter input that is SMU2 whereas a 0 Voltage was applied to the rest of circuit's pads (SMU1, SMU3 & SMU4) as shown in fig.1. During the measure step, a conventional DC configuration was applied, where a DC voltage source of 5V was applied by SMU2, a constant voltage of 5V by SMU1 VDD=5V and the output of the inverter was measured by

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SMU4. The switch between stress and measure configurations was automated via a LabVIEW program.

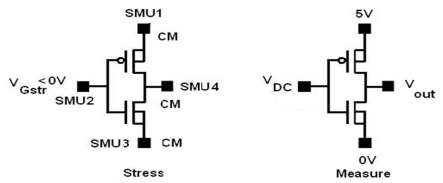

Fig.1 Measure-Stress-Measure protocol used for DC NBTI characterization

First, a measure of a virgin inverter's VTC has been done at ambient temperature, then the inverter has undergone a series of negative bias voltages of -8V, -10V, -12V and -13V for 1 hour where after each stress, the data of the VTC have been saved. This test protocol has been conducted for elevated temperatures ranging from 80°C to 140°C with a 20°C step using the hotplate in order to combine large field and elevated temperature for stress conditions.

III. RESULTS AND DISCUSSION

Fig. 2 shows the NBTI influence on the inverter voltage transfer curve VTC. It outcomes that NBTI causes a VTC shift to the left side as expected in the theory, that this VTC shift increases more and more with the increase of the applied field. This VTC shift induces a decrease of the logic threshold Vinv defined as Vout=Vin and found where the VTC crosses the X=Y line.

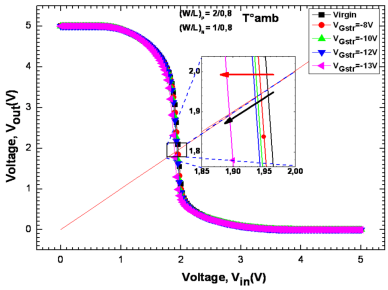

Fig. 2 Field influence on Inverter VTC under DC NBTI.

Fig. 3 shows the NBTI induced inverter VTC shift for a given stress voltage at different temperatures. It outcomes that the inverter VTC has the same trend of shift, and that the shift increases with elevated temperatures resulting in a decrease of the inverter logic threshold.

In order to demonstrate the influence of DC NBTI on the CMOS inverter, we have studied its impact on its main DC

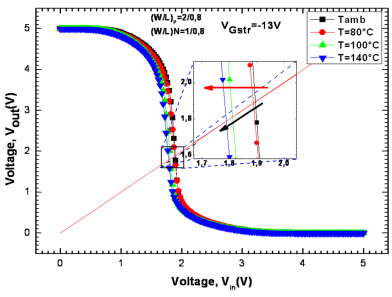

Fig. 3 Temperature influence on Inverter VTC under DC NBTI.

features which concern the logic threshold, the tolerable input and output logic levels, and its noise margins. These features qualify the robustness of the inverter and consequently its reliability. In the following subsections, we present these extracted features from the measured VTC under DC NBTI.

A. Inverter logic threshold

Fig.4.1 and fig 4.2 show the CMOS inverter logic threshold trend under DC NBTI as a function of the stress temperature

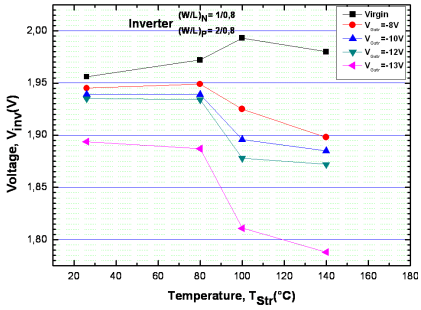

Fig. 4.1 Inverter logic threshold decrease with temperature under DC NBTI

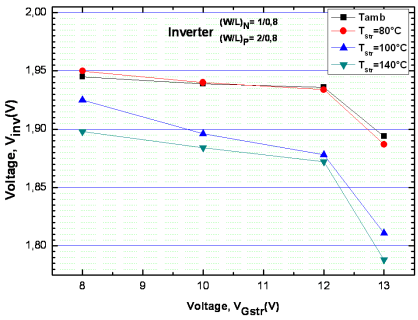

Fig. 4.2 Inverter logic threshold decrease with stress voltage under DC NBTI.

and stress voltage, respectively. Indeed, figure 4.1 shows that the logic threshold deceases with increasing temperature and the shift becomes more significant when the temperature is elevated above 80°C for a given stress voltage. We note here that logic threshold of fresh (unstressed) inverters noted by virgin in the figure has been improved and approaches the one of symmetrical VTC due to the annealing of the samples before stress.

The same shift trend of the inverter logic threshold is observed for increasing stress negative bias voltages as shown in fig 4.2 and it becomes more important when the applied field increases in its absolute value above 6.5 MV/cm corresponding to -13V for our samples of 20nm T_{ox}. Indeed, the logic threshold decreases more for negative stress voltage upper than 6MV/cm and worsen at elevated temperatures (more than 100°C).

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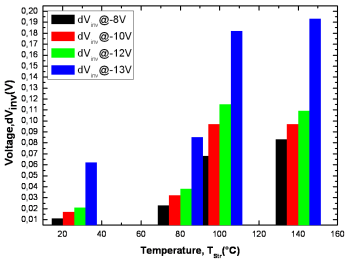


Fig. 5.1 Absolute shift of the inverter logic threshold under DC NBTI stress

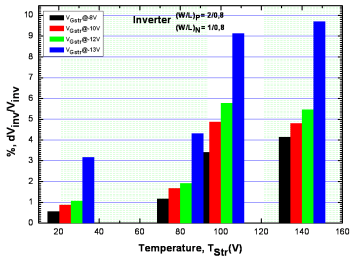


Fig. 5.2 Relative shift of the inverter logic threshold under DC NBTI stress.

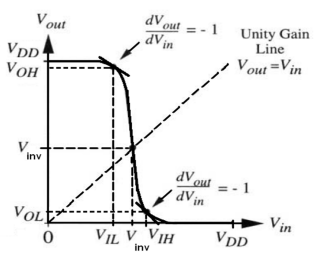
The DC NBTI induced shift of the inverter's logic threshold for different temperatures is given by the flowchart in fig 5.1.

It can be seen that the shift varies from 10mV to 60mV at ambient temperature and from 80mV to 195mV at 140°C. Accordingly, the relative shift caused by high negative stress voltage is more significant and varies from 3.1% at ambient temperature to more than 9.6% at 140°C (as depicted in flowchart of fig5.2).

This shift of the logic threshold implies unequal logic states of the inverter by enlarging the voltage range interpreted as logic "1" and shrinking the one for logic "0" as will be discussed in the following section.

B. *Logic voltage levels*

On behalf of the logic threshold, we have studied the influence of NBTI on the logic voltage levels that define the critical voltages of the CMOS inverter VTC to know V_{IL}, V_{IH}, V_{OL} and V_{OH} which are defined where the slope of the VTC has a value of -1 as shown in fig.6. Where, the input low voltage V_{IL} represents the largest value of Vin that can be interpreted as logic "0" whereas the input high voltage V_{IH} represents the smallest value of Vin that can be interpreted as



Fi.6. Critical CMOS inverter VTC voltages.

logic "1". The output low voltage V_{OL} defines the largest value of Vout's logic "0" while the output high voltage V_{OH} represents the smallest value of Vout's logic "1".

The values we extracted for these critical voltages from both fresh and stressed inverter for a series of temperatures and voltages are depicted in figure 7.

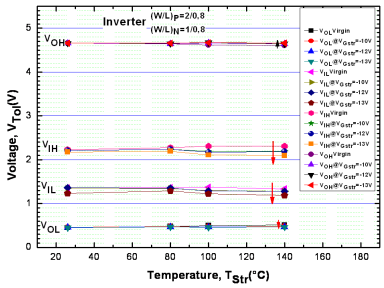


Fig.7 Measured logic voltage levels of the fresh and stressed CMOS inverters.

According to these results, we can say that NBTI has slightly influenced the CMOS inverter output levels V_{OL} and V_{OH} by decreasing the former and increasing the later. Consequently, the logic swing of the inverter defined as the difference between these two voltages is somewhat improved. In the other side, both input low and high voltages have been more affected and they have been decreased under NBTI stress. As a result, the input high voltage has been enlarged whereas its low voltage has been shrunk. Hence, the noise margins of the CMOS inverter will be obviously affected as will be demonstrated.

C. *Noise immunity*

The word *noise* in the context of digital circuits means "unwanted variations of voltages and currents at the logic nodes" [10]. This noise can be due to a capacitive, an inductive cross talk, or can be internally-generated power supply noise. Whatever its source, for a gate to be robust and insensitive to noise, it is essential that the "0" and "1" intervals be as large as possible. A measure of the inverter sensitivity to noise is given by the noise margins NML (noise margin low) and NMH (noise margin high), which quantize the size of the legal "0" and "1", respectively, and set a fixed maximum threshold on the noise value. So, we define the voltage noise margin for logic 1 (high) as:

$$NMH = V_{OH} - V_{IH} \quad (1)$$

Similarly, the voltage noise margin for logic 0 (low) as:

$$NML = V_{IL} - V_{OL} \quad (2)$$

Using these equations, we have computed the noise margins for fresh and stressed inverters and the results are depicted in fig.8. Fig 8.1 shows the influence of NBTI on noise margins with respect to temperature whereas fig 8.2 shows its influence as a function of the applied stress voltage. It is clear that both noise margins have been shifted under NBTI stress in opposite sides. While the noise margin high has been enhanced, the noise margin low has been decreased. As we are interested in NBTI induced noise margin degradation, we can see from fig 9.1 that the shift of NML caused by the application of a field of -6.5MV/cm (-13V) at 140°C is about 100mV representing a

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degradation of more than 10% of the fresh noise margin low. Thus, the noise margin at level "0" is seriously affected and also the inverter robustness.

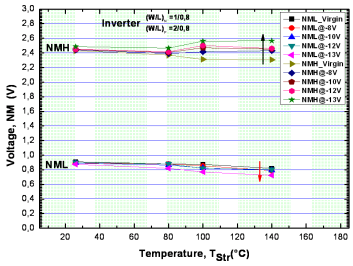

Fig.8.1 Measured inverter noise margins under NBTI stress with respect to temperature.

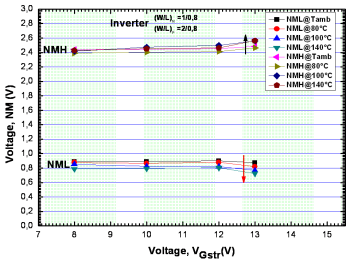

Fig.8.2 Measured inverter noise margins under NBTI stress with respect to stress voltage.

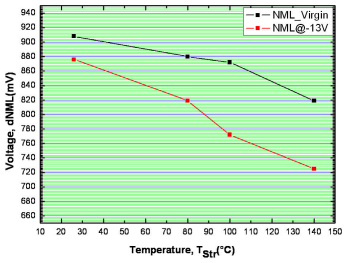

Fig.9.1 Absolute NML degradation under DC NBTI stress

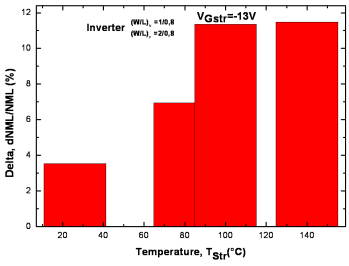

Fig.9.2 Relative NML degradation under DC NBTI stress.

IV. Conclusion

We have presented in this paper a deep investigation of NBTI impact on CMOS inverter DC performances. We have demonstrated that under NBTI, the inverter voltage transfer curve does shift to the left side as predicted by theory and that it experiences a more significant shift at higher negative voltage and worsen at elevated temperature. The results we obtained have demonstrated that all critical voltages have been affected. Indeed, the logic threshold has shifted and undergone a degradation of more than 10% of its nominal value after one hour of stress of -6.5MV/cm at 140°C. The input low and high voltages have been oppositely affected as the logic "1" has been enlarged whereas logic "0" has been shrunk leading to more unbalanced logic states. At the same time, the output voltages have been slightly affected by NBTI. Regarding inverter noise immunity, both noise margins have been influenced where the margin noise high NMH has been increased while the noise margin low has been shrunk. This later has undergone a degradation of more than 10% of its nominal value after one hour of stress of -6.5MV/cm and just at 100°C. Consequently, the CMOS inverter has become more vulnerable to noise, and less robust. Ultimately, this work has shown that CMOS inverter reliability has been seriously impacted by DC NBTI.

As PMOS transistors within digital circuits are not continuously experiencing a DC NBTI stress, circuit lifetime prediction upon DC NBTI stress is somewhat pessimistic. Accordingly, in future work, we will continue with the experimental investigation of AC NBTI impact on CMOS inverter temporal performances namely switching times and propagation delay to more evaluate both DC and AC NBTI at circuit level in order to develop an accurate model to perform NBTI ageing simulation needed by designers to evaluate their circuits lifetimes.

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